Wheat - Brown rust (leaf rust)

RAG record ADV-009 | canonical key: wheat_brown_rust

Record ID	ADV-009
Crop / crop key	Wheat / wheat
Condition / disease key	Brown rust (leaf rust) / brown_rust
Main pathogen	Puccinia triticina (older texts may use Puccinia recondita f. sp. tritici)
RAG aliases	wheat brown rust, wheat leaf rust, Puccinia triticina, Puccinia recondita

Farmer quick guide: treatments protect healthy plant tissue and slow disease spread. Dead or badly damaged leaf tissue will not become green again.

How to recognize it

- Look for orange-brown to brown powdery pustules, mainly scattered on the leaf blades.
- The pustules are usually less stripe-like than yellow rust.

Weather or field conditions that favour it

Rust can spread rapidly by wind when a susceptible variety and suitable weather occur together.

What to do now

- Confirm the rust type before treatment because brown rust and yellow rust have different field patterns.
- Use resistant varieties and scout regularly during the rust season.
- If brown rust is active, use a CIBRC/PPQS-listed label option and respect the waiting period.

Chemical treatment - only after the disease is confirmed

Chemical / formulation	Rate from cited source	How to use this information	Source
Propiconazole 25% EC	500 ml/ha in about 750 L water	PPQS/CIBRC wheat-rust guidance lists this for brown, yellow and black	[S1]

Chemical / formulation	Rate from cited source	How to use this information	Source
		rust; waiting period stated as 30 days.	
Mancozeb 75% WP	1.5-2.0 kg/ha in about 750 L water	PPQS guidance lists this for brown and black rust.	[S1]
Kresoxim-methyl 44.3% SC	500 ml/ha in about 500 L water	PPQS guidance lists this for wheat rust; waiting period stated as 25 days.	[S1]

Safety and compliance: Use only a product whose current Indian label allows the crop and target disease. Follow the label rate, water volume, re-entry interval and pre-harvest interval. Wear label-required PPE. Do not mix products unless the label or a qualified extension adviser permits it.

Prevention and integrated management

- Use resistant/tolerant varieties recommended for the zone.
- Avoid excessive nitrogen and overly dense crops.
- Monitor nearby fields because wind-borne spores can move quickly.

RAG guardrail

The AI should treat the chemical table as source-grounded examples, not permission to invent a different pesticide or dose. If the retrieved source does not match the crop, disease and formulation, return non-chemical advice and request local verification.

Sources used for this record

[S1] Directorate of Plant Protection, Quarantine & Storage (PPQS), Government of India. Wheat Rust - CIBRC-recommended pesticides and IPM guidance. https://ppqs.gov.in/sites/default/files/wheat_rust-english.pdf